6 **D** Taking direction downwards as positive,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$10 = -2.0t + \frac{1}{2}(9.81)t^2$$

$$t = -1.239 \text{ s (reject) or } 1.646 \text{ s}$$

$$\text{Range} = u_x t$$
$$= 5.0 \times 1.646$$
$$= 8.23 = 8.2 \text{ m}$$

7 **C** By principle of conservation of momentum, take direction to the right as positive for the